

Demonstration of Proposed Features

Date	2July,2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features

This document summarizes the features proposed during the project planning phase and verifies their successful implementation and demonstration in the **OptiCrop – Smart Agricultural Production Optimization Engine**.

Feature Demonstration

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	Soil Data Input	Accepts soil nutrient and environmental parameters through a user-friendly web interface.	Implemented	Yes	Successfully accepts user inputs.
2	Data Preprocessing	Validates and prepares user input for crop prediction.	Implemented	Yes	Ensures accurate model input.
3	Crop Recommendation	Predicts the most suitable crop using the trained Machine Learning model.	Implemented	Yes	Random Forest model integrated successfully.
4	Result Display	Displays the recommended crop along with image, season, water requirement, and cultivation tips.	Implemented	Yes	Provides detailed crop information instantly.
5	Responsive Web Interface	Provides a clean, responsive, and easy-to-use interface across devices.	Implemented	Yes	Smooth user experience.
6	Flask Web Application Deployment	Deploys the complete application on Render for online access.	Implemented	Yes	End-to-end workflow demonstrated successfully.

Feature Implementation Summary

Feature	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%

6	Flask Web Application Deployment	Integrate the trained model into a Flask-based web application for real-time use.	Implemented	Yes	Successfully demonstrated end-to-end workflow.
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Feature Implementation Summary

Feature	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate(%)	100%